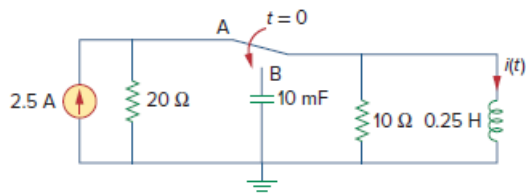


Problem

The switch in Fig. 16.49 moves from position A to position B at $t = 0$ (please note that the switch must connect to point B before it breaks the connection at A, a make before break switch). Determine $i(t)$ for $t > 0$. Also assume that the initial voltage on the capacitor is zero.

Figure 16.49:



Step-by-step solution

Step 1 of 6

Refer to Figure 16.49 in the text book.

Draw the circuit diagram at $t < 0$.

At $t < 0$, inductor acts as short circuit and capacitor acts as open circuit.

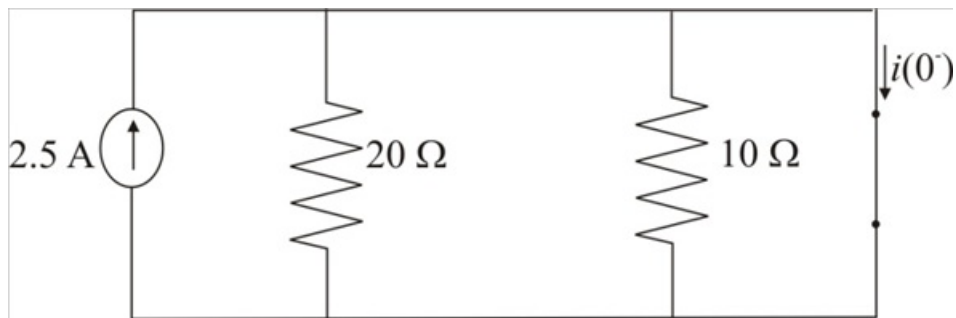
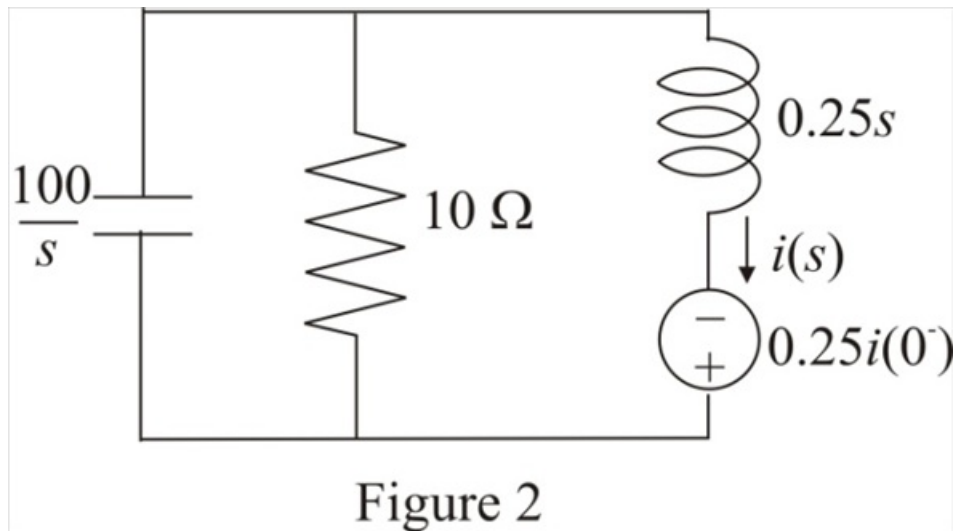


Figure 1

No current passes through the resistors since they are shorted. Therefore, the value of the current $i(0^-)$ is 2.5 A.

Step 2 of 6

Draw the circuit diagram at $t > 0$. Represent the circuit parameters in Laplace transform.



Step 3 of 6

Consider the following equation.

$$i(s) \left(0.25s + \left(\frac{100}{s} \right) \parallel (10) \right) - 0.25i(0^-) = 0$$

Substitute 2.5 A for $i(0^-)$.

$$i(s) \left(0.25s + \left(\frac{100}{s} \right) \parallel (10) \right) - 0.25(2.5) = 0$$

$$i(s) \left(0.25s + \frac{\left(\frac{100}{s} \right) (10)}{\frac{100}{s} + 10} \right) - 0.675 = 0$$

$$i(s) \left(0.25s + \frac{\left(\frac{1000}{s} \right)}{\frac{100 + 10s}{s}} \right) = 0.675$$

$$i(s) \left(0.25s + \frac{100}{s + 10} \right) = 0.675$$

$$i(s) \left(\frac{0.25s^2 + 2.5s + 100}{s + 10} \right) = 0.675$$

$$i(s) = \frac{0.675(s + 10)}{0.25s^2 + 2.5s + 100}$$

$$i(s) = \frac{2.5(s + 10)}{s^2 + 10s + 400}$$

$$i(s) = \frac{2.5(s + 10)}{(s + 5 - j19.365)(s + 5 + j19.365)} \dots\dots (1)$$

Step 4 of 6

Assume $\frac{2.5(s+10)}{(s+5-j19.365)(s+5+j19.365)}$ to be $\frac{A_1}{(s+5-j19.365)} + \frac{A_2}{(s+5+j19.365)}$.

$$\frac{2.5(s+10)}{(s+5-j19.365)(s+5+j19.365)} = \frac{A_1}{(s+5-j19.365)} + \frac{A_2}{(s+5+j19.365)}$$

Apply residue method to find the value of the constant A_1 .

$$\begin{aligned} A_1 &= (s+5-j19.365) \left(\frac{2.5(s+10)}{(s+5-j19.365)(s+5+j19.365)} \right) \Big|_{s=-5+j19.365} \\ &= \frac{2.5(s+10)}{(s+5+j19.365)} \Big|_{s=-5+j19.365} \\ &= \frac{2.5(-5+j19.365+10)}{-5+j19.365+5+j19.365} \\ &= \frac{2.5(5+j19.365)}{j38.72} \\ A_1 &= \frac{12.5+j48.4125}{j38.72} \\ &= 1.25-j0.323 \end{aligned}$$

Therefore, the value of the constant A_1 is $1.25-j0.323$.

Step 5 of 6

Apply residue method to find the value of the constant A_2 .

$$\begin{aligned} A_2 &= (s+5+j19.365) \left(\frac{2.5(s+10)}{(s+5-j19.365)(s+5+j19.365)} \right) \Big|_{s=-5-j19.365} \\ &= \frac{2.5(s+10)}{(s+5-j19.365)} \Big|_{s=-5-j19.365} \\ &= \frac{2.5(-5-j19.365+10)}{-5-j19.365+5-j19.365} \\ &= \frac{2.5(5-j19.365)}{-j38.72} \\ A_2 &= \frac{12.5-j48.4125}{-j38.72} \\ &= 1.25+j0.323 \end{aligned}$$

Therefore, the value of the constant A_2 is $1.25+j0.323$.

Step 6 of 6

Recall equation (1).

$$i(s) = \frac{2.5(s+10)}{(s+5-j19.365)(s+5+j19.365)}$$

Substitute $\frac{A_1}{(s+5-j19.365)} + \frac{A_2}{(s+5+j19.365)}$ for $\frac{2.5(s+10)}{(s+5-j19.365)(s+5+j19.365)}$.

$$i(s) = \frac{A_1}{(s+5-j19.365)} + \frac{A_2}{(s+5+j19.365)}$$

Substitute $1.25 + j0.323$ for A_2 and $1.25 - j0.323$ for A_1 .

$$\begin{aligned} i(s) &= \frac{1.25 - j0.323}{(s+5-j19.365)} + \frac{1.25 + j0.323}{(s+5+j19.365)} \\ &= 1.25 \left(\frac{1}{s+5-j19.365} + \frac{1}{s+5+j19.365} \right) + j0.323 \left(\frac{-1}{s+5-j19.365} + \frac{1}{s+5+j19.365} \right) \end{aligned}$$

Apply inverse Laplace transform.

$$\begin{aligned} i(t) &= 1.25 \left(e^{-(5-j19.365)t} + e^{-(5+j19.365)t} \right) + j0.323 \left(-e^{-(5-j19.365)t} + e^{-(5+j19.365)t} \right) \\ &= 1.25e^{-5t} \left(e^{j19.365t} + e^{-j19.365t} \right) + j0.323e^{-5t} \left(-e^{j19.365t} + e^{-j19.365t} \right) \\ &= 2.5e^{-5t} \left(\frac{e^{j19.365t} + e^{-j19.365t}}{2} \right) + 0.646e^{-5t} \left(\frac{e^{j19.365t} - e^{-j19.365t}}{j2} \right) \\ &= 2.5e^{-5t} \cos(19.365t) + 0.646e^{-5t} \sin(19.365t) \end{aligned}$$

$$\begin{aligned} i(t) &= e^{-5t} \left[2.5 \cos(19.365t) + 0.646 \sin(19.365t) \right] \\ &= e^{-5t} \left[\frac{2.5 \cos(19.365t) + 0.646 \sin(19.365t)}{\sqrt{2.5^2 + 0.646^2}} \right] \left(\sqrt{2.5^2 + 0.646^2} \right) \\ &= e^{-5t} \left[\frac{2.5 \cos(19.365t) + 0.646 \sin(19.365t)}{\sqrt{6.25 + 0.803}} \right] \left(\sqrt{6.25 + 0.803} \right) \\ &= e^{-5t} \left[\frac{2.5 \cos(19.365t) + 0.646 \sin(19.365t)}{\sqrt{7.053}} \right] \left(\sqrt{7.053} \right) \end{aligned}$$

$$\begin{aligned} i(t) &= e^{-5t} \left[\frac{2.5 \cos(19.365t) + 0.646 \sin(19.365t)}{2.656} \right] (2.656) \\ &= e^{-5t} \left[0.941 \cos(19.365t) + 0.243 \sin(19.365t) \right] (2.656) \end{aligned}$$

Let 0.941 be $\cos \theta$ so that 0.243 is $\sin \theta$ and θ is $\tan^{-1} \left(\frac{0.243}{0.941} \right)$

$$\begin{aligned} i(t) &= e^{-5t} (2.656) \left[\cos \theta \cos(19.365t) + \sin \theta \sin(19.365t) \right] \\ &= e^{-5t} (2.656) \cos(19.365t - \theta) \end{aligned}$$

Substitute $\tan^{-1} \left(\frac{0.243}{0.941} \right)$ for θ .

$$\begin{aligned} i(t) &= e^{-5t} (2.656) \cos \left(19.365t - \tan^{-1} \left(\frac{0.941}{0.243} \right) \right) \\ &= e^{-5t} (2.656) \cos(19.365t - \tan^{-1}(0.258)) \\ &= 2.656e^{-5t} \cos(19.365t - 14.47^\circ) \text{ A} \end{aligned}$$

Therefore, the value of the current $i(t)$ is $\boxed{2.656e^{-5t} \cos(19.365t - 14.47^\circ) \text{ A}}$.